

Week 3 Material - Differentiation of Vector-Valued Functions

11. ¹ (a) Define a vector-valued function for the path of a particle that:
- begins at $(x, y) = (2, 0)$ at $t = 0$;
 - moves clockwise around a circle of radius 2 (centered at the origin); and
 - moves at a constant speed of 3 m/s. (Assume t is in seconds, and x and y are in meters.)
- (b) Verify that your answer to part (a) satisfies the speed requirement by finding a general formula for the velocity of the particle, and then the resulting speed.
- (c) Find at least one time value at which this particle would be moving in a direction parallel to the vector $\langle -2, 2 \rangle$.
12. ² Design a 2D vector-valued function $\vec{s}(t) = \langle x(t), y(t) \rangle$ that describes a trajectory which:
- Follows a radius 10 circle around the origin;
 - Starts at position $(-10, 0)$ at $t = 0$;
 - Moves clockwise at a constant speed of 5 m/s.
- Assume all distances are in meters and time is measured in seconds.
13. (a) Sketch the plane curve with vector equation $\mathbf{r}(t) = \langle t - 2, t^2 + 1 \rangle$.
- (b) Sketch the position vector $\mathbf{r}(t)$ and the tangent vector $\mathbf{r}'(t)$ at $t = -1$.
14. ³ Tenderheart Bear, one of the mystical Care Bears™, is driving in his Cloud Car, with his position given by

$$\mathbf{r}(t) = \langle t + 1, t^2 + 2 \rangle, \quad -\infty < t < \infty$$

His friend Funshine Bear is standing at coordinates $(-1, 2)$.

When Tenderheart presses a button in his Cloud Car, the car will send a magical ‘hug’ straight ahead in the direction the car is moving at that moment. If Tenderheart wants to send a hug to Funshine without having to stop, at what time t should Tenderheart press the button?

¹Source: previous APSC 171 Test 1

²Source: previous APSC 171 Test 1

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